Wall time per step: Splat simulation

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

100k (96,342 cells)

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells
load	8.4	5.8	6.0	5.5	18	2.3	5.1	34	6.1	47	38	9.4	13
qc	0.1	0.1	0.1	0.1	0.6	0.6	0.9	0.5	0.5	1.4	2.6	2.7	1.1
normalise	n/a	n/a	n/a	n/a	2.0	1.9	0.0	0.1	0.1	0.0	0.2	0.2	0.4
hvg	0.6	0.3	0.4	0.2	1.4	1.4	1.4	0.1	0.1	3.8	3.7	3.8	2.9
pca	0.8	0.8	0.8	0.6	2.2	2.2	3.9	1.8	1.8	5.1	11	11	58
harmony	0.5	0.5	0.8	0.3	0.5	0.5	0.5	12	14	14	13	13	14
knn_snn	2.6	2.8	1.6	1.4	27	27	24	6.4	6.4	6.8	7.5	7.6	7.8
cluster	0.9	1.4	0.8	0.8	3.0	2.7	2.7	13	14	13	19	19	21
cluster_fast	1.3	1.0	0.4	0.5	n/a	n/a	n/a	18	18	17	n/a	n/a	n/a
umap	3.7	3.7	0.8	0.7	39	38	38	43	44	43	46	46	46
dge	0.5	0.5	0.5	0.5	47	46	–	3.4	3.5	79	12	11	89
tsne_default	44	43	36	33	162	167	155	n/a	n/a	n/a	125	122	147
tsne_fft	15	15	2.8	2.8	38	38	36	n/a	n/a	n/a	32	32	33
tsne_quick	11	11	3.8	3.4	n/a	n/a	n/a	25	25	23	n/a	n/a	n/a
total (no tSNE)	18	16	12	10	141	122	77	115	90	213	152	124	253

500k (482,409 cells)

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells
load	49	32	28	23	OOM	11	21	205	23	224	225	41	66
qc	0.3	0.2	0.3	0.3	–	2.8	3.8	2.2	2.2	6.3	14	13	4.0
normalise	n/a	n/a	n/a	n/a	–	3.6	0.0	0.4	0.4	0.0	0.9	0.8	1.9
hvg	0.7	0.8	0.7	0.7	–	3.1	4.5	0.5	0.5	19	19	19	13
pca	2.4	2.7	1.5	1.1	–	10	11	14	14	31	OOM	OOM	271
harmony	1.6	1.8	1.2	0.7	–	2.1	2.0	100	84	82	–	–	88
knn_snn	13	14	7.2	6.4	–	33	29	40	40	40	–	–	45
cluster	3.9	4.2	3.4	3.2	–	20	20	116	110	102	–	–	132
cluster_fast	14	14	3.1	2.4	n/a	n/a	n/a	196	194	195	n/a	n/a	n/a
umap	35	35	3.7	3.5	–	226	226	213	219	213	–	–	256
dge	1.8	1.9	1.7	1.7	–	454	–	22	22	1,710	–	–	404
tsne_default	261	256	232	252	–	1,315	1,319	n/a	n/a	n/a	–	–	1,404
tsne_fft	65	64	14	13	–	153	156	n/a	n/a	n/a	–	–	157
tsne_quick	44	43	26	25	n/a	n/a	n/a	155	152	153	n/a	n/a	n/a
total (no tSNE)	108	93	47	40	failed	766	317	712	515	2,426	failed	failed	1,281

1M (964,763 cells)

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells
load	101	88	69	45	OOM	OOM	39	OOM	OOM	415	OOM	OOM	133
qc	0.5	1.5	0.4	0.4	–	–	7.2	–	–	13	–	–	9.8
normalise	n/a	n/a	n/a	n/a	–	–	0.0	–	–	0.0	–	–	4.3
hvg	1.4	1.6	1.3	1.1	–	–	8.2	–	–	38	–	–	25
pca	5.1	5.4	2.6	2.0	–	–	18	–	–	68	–	–	468
harmony	3.2	3.1	2.0	1.2	–	–	4.2	–	–	170	–	–	179
knn_snn	28	30	15	14	–	–	40	–	–	92	–	–	100
cluster	8.3	8.9	6.7	6.4	–	–	47	–	–	275	–	–	367
cluster_fast	46	46	8.0	6.3	n/a	n/a	n/a	–	–	555	n/a	n/a	n/a
umap	81	77	10	9.8	–	–	555	–	–	436	–	–	555
dge	3.2	3.3	5.5	3.0	–	–	–	–	–	TIMEOUT	–	–	822
tsne_default	658	623	557	590	–	–	TIMEOUT	n/a	n/a	n/a	–	–	TIMEOUT
tsne_fft	206	193	29	40	–	–	–	n/a	n/a	n/a	–	–	–
tsne_quick	140	134	96	80	n/a	n/a	n/a	–	–	–	n/a	n/a	n/a
total (no tSNE)	232	218	112	83	failed	failed	719	failed	failed	failed	failed	failed	2,662

per step: fastest … slowest

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. bixverse GPU and MLX run natively without a memory cap.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.